

AI TOOLS THAT WORK

Check the Math

Three everyday calculations: a recipe, a loan and a phone plan. Checked with a calculator.

Tested October 4, 2026

Why this guide

Three small number jobs: scale a pancake recipe for a party, work out a loan payment before signing, compare two phone plans. None of it is hard, but a slip costs real money or a flat pancake.

I asked an AI chat to do the arithmetic, then checked each answer a second way. The check takes a minute.

The numbers are made up for practice. They are not real offers. The AI answers and my checks are real.

How I tested

I used a free public chat demo, with no account and no payment: the Qwen3 demo on Hugging Face

(huggingface.co/spaces/Qwen/Qwen3-Demo), model shown as Qwen3-235B-A22B.

I ran three chats on October 4, 2026, from about 12:09 to 12:14 AM Saudi time, one per example. Only invented practice numbers went in. I checked each answer with my own calculations in Python, which you can repeat with a phone calculator.

All three came out right on the numbers this time. One run each, so another run, or another day, can give a different answer.

Keep it private. This is a public demo. Do not paste real account numbers, loan documents or statements.

EXAMPLE 1

The pancake party

Your recipe makes 8 pancakes. You need 20. Scaling by hand means fractions of cups and teaspoons, which is where slips happen.

I pasted the recipe and asked for the multiplier and the new amounts.

My check: 20 divided by 8 is 2.5. Each amount times 2.5.

What the AI wrote

ANSWER, 22.5 SECONDS

Multiplier: 2.5

Flour: 3.75 cups

Sugar: 2.5 tablespoons

Baking powder: 8.75 teaspoons

Milk: 3.125 cups

Egg: 2.5

Melted butter: 7.5 tablespoons

What it got right

- The multiplier is 2.5. Correct.
- Every amount matches my check: 3.75 cups flour, 2.5 tablespoons sugar, 8.75 teaspoons baking powder, 3.125 cups milk, 7.5 tablespoons butter.

What needed fixing

- "Egg: 2.5" is correct arithmetic and useless in a kitchen. It did not say to round or how to split an egg. Decide yourself: 2 eggs, or beat 3 and use $\frac{5}{6}$ of that.
- 8.75 teaspoons is awkward. 3 teaspoons make a tablespoon, so that is about 2 tablespoons and $2\frac{3}{4}$ teaspoons. I did not ask it to simplify, and it did not.

EXAMPLE 2

The loan

You are offered 12000 dollars at 7.5 percent a year for 36 months. You want the monthly payment and the total interest.

This is a standard formula, so I asked the AI to show the formula and the steps.

My check

Monthly rate: 7.5 divided by 12 is 0.625 percent.

Payment = $P \times r / (1 - (1 + r)^{-n})$ to the power of minus n). I ran it in Python:

payment = 373.2746 dollars

total paid = $373.2746 \times 36 = 13437.89$ dollars

interest = 1437.89 dollars

What the AI wrote

ANSWER, 194.6 SECONDS (ABOUT 3 MINUTES)

Step 1: Calculate $(1 + r)^n$

$(1 + 0.00625)^{36} = 1.251446$ (approximately)

Step 2: Plug into the formula

$M = 12000 \times 0.00625 \times 1.251446 / (1.251446 - 1)$

$M = 12000 \times 0.0078215375 / 0.251446$

$M = 12000 \times 0.031106 = 373.27$ (approximately)

Step 3: Calculate total interest paid

Total payments = $373.27 \times 36 = 13437.72$

Total interest = $13437.72 - 12000 = 1437.72$

Final Answer

The monthly payment is 373.27 dollars and the total interest paid is 1437.72 dollars.

Tidied for reading: the demo split the formula into one symbol per line, so I rebuilt the steps from the same

numbers. Nothing in the figures changed.

Checking the loan

What it got right

- Monthly payment: 373.27 dollars. Matches my 373.2746, rounded.
- The formula and steps are the standard ones, and $(1 + 0.00625)$ to the power 36 is about 1.251446, as it says.

What needed fixing

- Total interest: it says 1437.72 dollars. I get 1437.89. The 17 cent gap is rounding: it multiplied the rounded 373.27 by 36. A lender may round differently, so real figures can differ by cents. Ask your lender for the exact schedule.
- It took about 3 minutes and a long thinking section before the answer. A one line formula in a spreadsheet is faster.
- In its visible thinking it wandered and tried a rougher value (1.2514) before settling on 1.251446. The final answer was fine, but a rambling working-out is a reason to verify.

Spreadsheet check

In any spreadsheet: `=PMT(0.075/12, 36, -12000)`

That returns the monthly payment. Multiply by 36 and subtract 12000 for the interest.

EXAMPLE 3

Two phone plans

Say you are comparing two plans over 24 months. The numbers are invented for practice. Ignore taxes.

Plan A: 45 dollars a month, plus 200 dollars for the phone up front.

Plan B: 35 dollars a month, plus a 12 dollars a month device fee for 24 months, plus a one-time 50 dollars activation fee.

My check: A is $45 \times 24 + 200 = 1280$. B is $(35 + 12) \times 24 + 50 = 1178$. B is 102 cheaper.

What the AI wrote

ANSWER, 16.7 SECONDS

Plan A: Monthly payment of 45 dollars over 24 months equals 1080 dollars ($45 * 24$) plus a 200 dollars phone fee, totaling 1280 dollars.

Plan B: Monthly payment of 35 dollars plus 12 dollars device fee (total 47 dollars monthly) over 24 months equals 1128 dollars ($47 * 24$) plus a one-time 50 dollars activation fee, totaling 1178 dollars.

Plan B costs less by 102 dollars ($1280 - 1178$).

Total for Plan A: 1280 dollars. Total for Plan B: 1178 dollars. Plan B saves 102 dollars.

What it got right

- Both totals and the 102 dollars difference match my check.
- It showed each step, so each number can be checked.

What needed fixing

- It calls the 200 dollars a "phone fee". It is the price of the phone. Words matter if you copy this into a note.
- It does not say what the comparison leaves out: the plans may include different data, a different phone, or cancellation costs. The math is right, but cheaper is not the same as better.

What I took from this

All three answers were right on the numbers. The problems were around the numbers: half an egg, 17 cents of rounding, a loose label. The arithmetic may be fine, and you still check it, and you still read what it left out.

Before you rely on a number an AI gave you:

1. Is every number in the answer one you gave it?
2. Did it show the steps, so you can redo one?
3. Rerun it on a calculator or spreadsheet.
4. Does the last digit move when you round differently?
5. Can you use the answer in real life? Half an egg, for example.

The prompts

Copy these and change the numbers to match your situation.

PROMPT 1: SCALE A RECIPE

A pancake recipe makes 8 pancakes. Ingredients: 1.5 cups flour, 1 tablespoon sugar, 3.5 teaspoons baking powder, 1.25 cups milk, 1 egg, 3 tablespoons melted butter.

Scale the recipe to make 20 pancakes. Show the multiplier, then list each ingredient with its new amount. Use cups, tablespoons and teaspoons as in the original. Plain text only, no tables.

PROMPT 2: LOAN PAYMENT

Sam borrows 12000 dollars at 7.5 percent annual interest for 36 months, with fixed monthly payments and monthly compounding. What is the monthly payment, and how much total interest does Sam pay over the whole loan? Show the formula and the steps. Plain text only, no tables. Write money as N dollars.

PROMPT 3: COMPARE TWO PLANS

Sam is comparing two phone plans over 24 months. Ignore taxes.

Plan A: 45 dollars per month, and Sam pays 200 dollars for the phone up front.

Plan B: 35 dollars per month, plus a 12 dollars per month device fee for 24 months, plus a one-time 50 dollars activation fee.

Which plan costs less over 24 months, and by how much? Show the total for each plan. Plain text only, no tables. Write money as N dollars.

Limits and sources

- One public demo model, one day, one run per prompt. Another run or another tool can give a different answer, including a wrong one.
- I did not test on a phone.
- The loan formula is the standard fixed-payment amortization formula. Real loans can add fees or round differently.
- The numbers in the examples are invented. They are not offers.
- This is not financial advice. Check real terms with your lender or provider.

SOURCES

Qwen3 demo:

huggingface.co/spaces/Qwen/Qwen3-Demo

My checks: Python 3 and standard loan amortization formula.

More free guides from AI Tools That Work

All ten are free to download here:

faiz12321.github.io/ai-tools-that-work

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